

Influence of electronF2 collisions in rare gas–halide laser discharges

William L. Nighan

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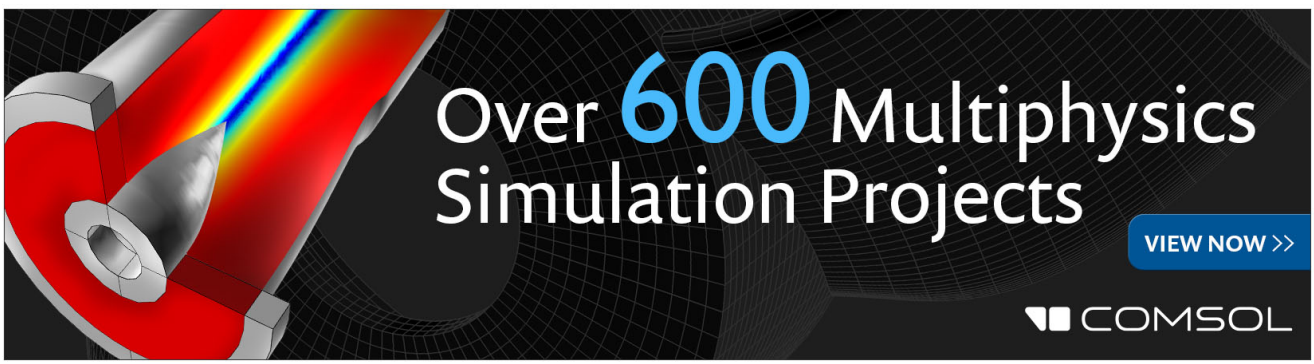
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[1]For a convenient review, see: R. Dingle, in *Festkorperprobleme XV (Advances in Solid State Physics)*, edited by H.J.Queisser (Pergamon-Vieweg, Berlin, 1975), pp. 21—48.
[2]MBE is reviewed by A.Y. Cho and J.R. Arthur, Jr., in *Progress in Solid State Chemistry*, edited by J.O. McCaldin and G.A. Somorjai (Pergamon, New York, 1975), Vol. 10, pp. 157—191.
[3]E.A. Rezek, N. Holonyak, Jr., B.A. Vojak, G.E. Stillman, J.A. Rossi, D.L. Keune, and J.D. Fairing, Appl. Phys. Lett. **31**, 288 (1977).
[4]E.A. Rezek, H. Shichijo, B.A. Vojak, and N. Holonyak, Jr., Appl. Phys. Lett. **31**, 534 (1977).
[5]E.A. Rezek, N. Holonyak, Jr., B.A. Vojak, and H. Shichijo, J. Appl. Phys. **49**, 69 (1978).
[6]R.D. Dupuis and P.D. Dapkus, Appl. Phys. Lett. **31**, 466 (1977).
[7]R.D. Dupuis and P.D. Dapkus, Appl. Phys. Lett. **31**, 839 (1977).
[8]R.D. Dupuis, P.D. Dapkus, and L.A. Moudy, IEEE Int. Electron Device Meeting, Washington, D.C., 1977 (unpublished).
[9]R.D. Dupuis, P.D. Dapkus, R.D. Yingling, and L.A. Moudy, Appl. Phys. Lett. **31**, 201 (1977).
[10]The doping levels are assumed from values measured by the van der Pauw technique at room temperature on thicker (8—10 μm) films grown under the same conditions on (100) GaAs:Cr substrates. Layer thicknesses are measured by SEM examination of cleaved cross sections that have been etched with H_2O_2 (30%) adjusted to *p*H 8.02 with NH_4OH.
[11]B. Schwartz, J.C. Dyment, and S.E. Haszko, *Proceedings 4th International Symposium on GaAs and Related Compounds*, Boulder, 1972 (Institute of Physics and Physical Society, London, London, 1973), p. 187.
[12]L. Ralph Dawson, J. Appl. Phys. **48**, 2485 (1977).
[13]R. Ludeke, L. Esaki, and L.L. Chang, Appl. Phys. Lett. **24**, 417 (1974).
[14]J.P. van der Ziel, R. Dingle, R.C. Miller, W. Wiegmann, and W.A. Nordland, Jr., Appl. Phys. Lett. **26**, 463 (1975).
[15]R.C. Miller, R. Dingle, A.C. Gossard, R.A. Logan, W.A. Nordland, Jr., and W. Wiegmann, J. Appl. Phys. **47**, 4509 (1976).
[16]N. Holonyak, Jr., B.A. Vojak, E.A. Rezek, and R. Chin (unpublished).

Influence of electron-F_2 collisions in rare gas–halide laser discharges[a)]

William L. Nighan

United Technologies Research Center, East Hartford, Connecticut 06108
(Received 23 September 1977; accepted for publication 20 December 1977)

The influence of F_2 vibrational excitation and direct dissociation by electron impact is examined for conditions typical of electron-beam-sustained KrF* lasers. For values of F_2 fractional concentration greater than about 0.003 the results of this analysis indicate that the dissociation process $e + F_2 \rightarrow 2F + e$ may significantly affect electron-metastable atom production efficiency, rare gas–halide excimer production efficiency, and gain.

PACS numbers: 42.55.Hq, 52.20.Fs, 52.20.Hv

Rare gas—halide laser mixtures generally contain fractional concentrations of the halogen bearing molecule in the 0.1—1.0% range. Although there exists almost no electron scattering data for such molecules, there is every reason to suppose that the presence of several tenths of a percent F_2, for example, will have an effect on electron kinetics other than by way of the well-known dissociative attachment process. This paper analyzes available data relevant to electron-F_2 scattering and examines the potential importance of such collision processes for conditions typical of discharge-pumped KrF* lasers.

To date the effect of electron-halogen molecule inelastic collisions has not been considered in analyses of rare gas—halide lasers. This is a reflection of the lack of required cross-section data. However, recent theoretical analysis[1] has shown that vibrational excitation of F_2 by electrons can proceed by way of a resonance mechanism involving the same F_2^- state which participates in the low-energy dissociative attachment process. The computed energy weighted cross section for vibrational excitation was found to be in excess of 10^{-17} cm^2 in the electron energy range of a few electron volts. In addition, measurements[2] of F atom production in an electron-beam-sustained F_2 discharge have shown evidence of enhanced F_2 dissociation as E/n was increased for average electron energies estimated to be about 1 eV, a value high enough[1] so that dissociative attachment should be a decreasing function of E/n. While there are several processes[2] which may contribute to F_2 dissociation for the experimental conditions of Ref. 2, direct electron impact dissociation is by far the most likely. Indeed, there are several F_2^- states which may lie above the repulsive $F_2(^1\pi_u)$ state in the vicinity of the F_2 ground-state equilibrium separation, so that resonant enhancement of the process $e + F_2 \rightarrow 2F + e$ is a distinct possibility.

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In order to obtain an estimate for the electron-F_2 dissociation cross section based on the observations of Ref. 2, electron energy distributions were computed for F_2 using the vibrational cross sections of Ref. 1 along with a constant momentum transfer cross section having a value of 10^{-15} cm^2. The electron-F_2 dissociation cross section was assumed to have an apparent threshold of 3.35 eV corresponding to the vertical transition energy from the F_2 ground state to the $F_2(^1\pi_u)$ state, the lowest electronic state of F_2. The magnitude and shape of this cross section were then varied in a trial and error fashion. In the experiments of Ref. 2 the production of F atoms was first observed to increase over that due to the electron beam alone for an E/n value of approximately 2×10^{-16} V cm^2. As E/n was increased to about 6×10^{-16} V cm^2, F atom production was enhanced by almost a factor of 2. By using a trial dissociation cross section having a peak value of approximately 10^{-17} cm^2 at an electron energy of 5 eV, in the present work it was found that the magnitude of the calculated rate of F_2 dissociation (F atom production) by the direct process was a few percent of that due to dissociative attachment for an E/n of 2×10^{-16} V cm^2. Further, the dissociation rate approximately equaled the dissociative attachment rate for E/n values in the 5×10^{-16} to 6×10^{-16}-V cm^2 range, results consistent with the experimental observations.[2] These findings were not particularly sensitive to the shape of the trial cross section used in the analysis. The direct electron-F_2 dissociation rate estimated in this manner is consistent with the limited experimental data available[2] and provides a reasonable basis for an evaluation (albeit provisional) of the potential significance of electron impact excitation of F_2 to repulsive states.

Using this F_2 dissociation cross section and the cross sections of Ref. 1 the electron fractional power transfer

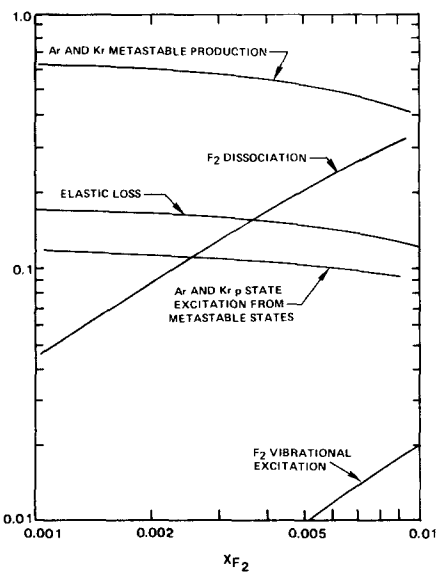


FIG. 1. Contributions to electron fractional power transfer in an Ar-Kr-F_2 mixture in which the Ar and Kr fractional concentrations were 0.95 and 0.05, respectively. The calculation was carried out for an E/n value 1.0×10^{-16} V cm^2, a fractional metastable concentration of 10^{-5}, and a fractional ionization of 10^{-6}.

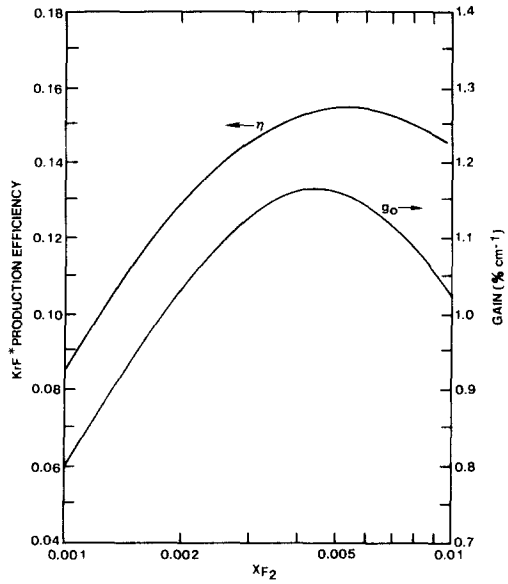


FIG. 2. Small-signal gain and KrF* production efficiency in an electron-beam-sustained discharge as a function of F_2 fraction. The computation was made for an atmospheric-pressure Ar-Kr (0.95—0.05) mixture and an E/n value of 1.0×10^{-16} V cm^2. These results refer to conditions 100 nsec after discharge initiation.

was computed for conditions typical of electric-discharge-pumped KrF* lasers.[3] The results of this calculation are presented in Fig. 1 as a function of F_2 fractional concentration. For values of X_{F_2} below about 0.002 electron energy loss due to dissociation is found to be relatively unimportant. However, for higher F_2 concentrations the electron energy loss accompanying dissociation can become very significant, exceeding 10% of the total discharge power. For the conditions of Fig. 1 the rate coefficient for direct dissociation has a value of approximately 10^{-9} sec^{-1} cm^3 and exhibits a weak dependence on E/n. Since the production of rare gas metastable states is a strong positive function of E/n, the fractional power transfer associated with the direct dissociation process decreases relative to rare gas metastable production for E/n values higher than 1.0×10^{-16} V cm^2 and increases for lower values.

Although the cross sections for F_2 vibrational excitation are relatively large,[1] Fig. 1 shows that the electron energy loss due to vibrational excitation is not likely to be important since the energy loss per collision is small (~0.1 eV). However, Hall's results[1] indicate that dissociative attachment increases significantly with F_2 vibrational level. In long-pulse[3] KrF* laser discharges the fraction of vibrationally excited F_2 will be substantial (>10%). Thus, the degree of F_2 vibrational excitation may affect the electron loss due to attachment, thereby influencing both the quasisteady and stability characteristics of rare gas—halide discharges.

In order to evaluate the influence of the electron-F_2 collision processes discussed above, electron distribution functions and all related Ar and Kr rate coefficients were computed as a function of F_2 fractional concentration. The effects of electron-electron, electron-ion, and electron-metastable collisions were taken into

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account in the calculation. The data so obtained were then used as input information in a time-dependent KrF* kinetics model of an electron-beam-sustained discharge,[3] following procedures generally similar to those described elsewhere.[4,5] Figure 2 shows the computed variation of gain and KrF* production efficiency with F_2 fraction. The results presented refer to conditions 100 nsec after discharge initiation. In the present analysis the KrF* production efficiency is defined by the relation $\eta = (\text{KrF* production})h\nu/([\text{Ar}]S_E u_i + JE)$, where all the processes contributing to the *volumetric* rate of KrF* production are included in the numerator.[6] Also, S_E is the rate of electron-ion pair production by the electron beam, u_i is the energy required to produce an electron-ion pair, JE is the discharge power density, and $h\nu$ is the photon energy of the KrF* laser transition. Thus, on this basis η represents the maximum fraction of the total power potentially recoverable from the reaction, $\text{KrF}^* + h\nu \rightarrow \text{Kr} + \text{F} + 2h\nu$.

Interpretation of the specific nature of changes accompanying variation in F_2 concentration can be very difficult for the circumstances typical of most experiments since the halogen molecules influence numerous plasma processes. Therefore, in order to assist in interpretation, in the present calculation the e-beam ionization rate, S_E, was varied from 50 to 500 sec^{-1} as the F_2 fraction was increased from 0.001 to 0.01. Thus, the electron density remains approximately constant at 10^{14} cm^{-3} for the conditions of Fig. 2. In addition, the *total* power density is sensibly constant, increasing from about 60 kW cm^{-3} to 90 kW cm^{-2} over the X_{F_2} range covered.[7] These conditions are representative of recent experiments[3] with electron-beam-sustained rare gas—halide lasers.

As the F_2 fraction is increased above 0.001, Fig. 2 indicates that both the gain and KrF* production efficiency increase significantly. This reflects the improved efficiency of ArF* formation from the rare gas metastable states as compared to other metastable loss processes. However, as X_{F_2} is increased above about 0.005, electron energy loss due to F_2 dissociation becomes increasingly more significant. The resultant cooling of the electrons leads to a 35% decrease in the Ar and Kr metastable production rates for the conditions of this example as X_{F_2} is increased from 0.001 to 0.01. The combined effect of these related processes is a reduction in the KrF* production efficiency.[8] Additionally, F_2 quenching of KrF* exerts an important deleterious influence on the gain at the higher F_2 levels.[9] Results qualitatively similar to those of Fig. 2 were obtained at lower and higher E/n values, although calculations show that the plasma is unstable for E/n values above about 1.2×10^{-16} V cm^2 when the F_2 fractional concentration is below about 0.002.

Although the dominant collision processes generally become quasisteady in times much less than 100 nsec for conditions typical of those represented by the results of Fig. 2, the temporal changes in plasma processes resulting from F_2 dissociation can become very significant. The consequences of F_2 dissociation become particularly important for long-pulse discharges (~1 μsec) and/or high-power loading (>100 kW cm^{-3}).

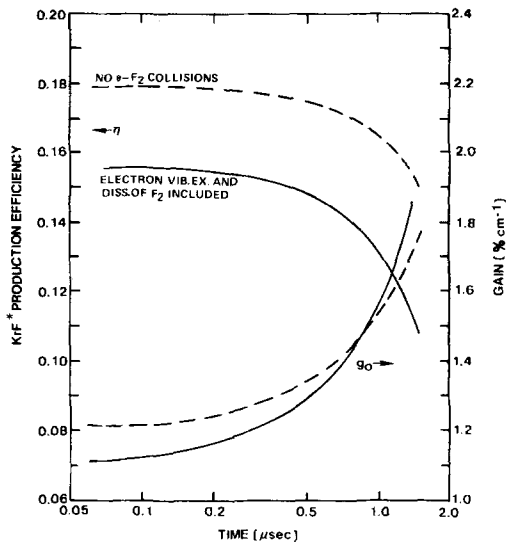


FIG. 3. Temporal variation of KrF* production efficiency and gain for an F_2 fraction of 0.005 and the conditions of Fig. 2. Also shown (dashed curves) are the results of the same calculation neglecting all electron-F_2 collisions except attachment.

Presented in Fig. 3 are the temporal variations of η and g_0 computed for an X_{F_2} value of 0.005, i.e., near-optimum conditions in Fig. 2. For purposes of comparison, also shown are η and g_0 computed neglecting the effects of all electron-F_2 collisions except attachment. Clearly, steady-state conditions are attained in a time less than 100 nsec. However, for times greater than about 0.2 to 0.3 μsec significant changes in both g_0 and η resulting from F_2 dissociation become apparent. The decrease in KrF* production efficiency is a sequence of the increase in electron density accompanying the decrease in F_2 concentration. As the electron density increases, electron excitation of rare gas metastables to higher excited states begins to compete with the desired F_2 reaction leading to the formation of ArF* and KrF*. For this reason the energy utilization efficiency associated with conversion of metastable atoms to rare gas—halide excimers decreases. Note, however, that because of increasing electron excitation, the absolute densities of metastable and rare gas—halide states continue to increase as is reflected by the increasing gain. As the data of Fig. 3 indicate, this sequence of events results in rapidly changing g_0 and η for long discharge duration. At the 1-μsec time the F_2 is 35% dissociated for the conditions of this example.

Comparison of the dashed and solid curves in Fig. 3 shows that direct electron impact dissociation of F_2 may result in significant changes in the magnitude and rate of change of both gain and KrF* production efficiency. However, it is clear that the general trends exhibited by the data in Fig. 3 are not dependent on the inclusion of electron-F_2 collisions. Indeed, at the 1-μsec time only about one-third of the F_2 dissociation results from direct electron impact for the conditions of this example, with ArF* formation, attachment, and F_2 quenching processes accounting for the other two-thirds.

The results and discussion of this paper focus attention on the role of the halogen-bearing molecule as

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regards electron and metastable kinetic processes in rare gas—halide lasers. Conditions typical of electron-beam-sustained laser discharges[3] were examined in detail. The influence of F_2 in KrF* lasers was given particular attention because of the availability (albeit limited) of theoretical and experimental data for this molecule. On the basis of this analysis it is reasonable to conclude that electron-halogen processes other than attachment are likely to be important, particularly F_2 electronic excitation leading to dissociation. This finding will most probably have general significance for all rare gas—halide and mercury-halide lasers containing F_2 or other halogenated molecules at fractional concentration levels greater than a few tenths of a percent. Clearly, additional electron scattering data for halogenated molecules would be of considerable value in the modeling and evaluation of rare gas—halide laser properties.

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[1]R.J. Hall, J. Chem. Phys. (to be published).
[2]H.L. Chen, R.E. Center, D.W. Trainor, and W.I. Fyfe, J. Appl. Phys. 48, 2297 (1977).
[3]C.H. Fisher and R.E. Center, Appl. Phys. Lett. 31, 106 (1977).
[4]J.H. Jacob and J.A. Mangano, Appl. Phys. Lett. 28, 724 (1976).
[5]J.A. Mangano, J.H. Jacob, and J.B. Dodge, Appl. Phys. Lett. 29, 426 (1976).
[6]For the conditions of this example the displacement reaction $ArF^* + Kr \rightarrow KrF^* + Ar$ dominates KrF* production (see Ref. 9).
[7]Since the discharge contribution to the total power density was sensibly constant over the entire X_{F_2} range covered in Fig. 2, the discharge enhancement factor, $JE([Ar]u_i S_E)^{-1}$ decreased from about 10 to 1 as X_{F_2} increased from 0.001 to 0.01, reflecting the increase in S_E required to maintain nearly constant electron density. However, the discharge contribution to the total volumetric power density was dominant over most of the range covered in Fig. 2.
[8]Due to the decrease in enhancement factor corresponding to the increase in X_{F_2} for the conditions described here, the discharge is relatively less important at high F_2 fractions; this effect also contributes to the reduction in η.
[9]M. Rokni, J.H. Jacob, and J.A. Mangano, Phys. Rev. (to be published).

An optical waveguide TE-TM mode splitter

Morio Kobayashi, Hiroshi Terui, and Kazumichi Egashira

Ibaraki Electrical Communication Laboratory, Nippon Telegraph and Telephone Public Corporation, Tokai, Ibaraki-ken 319-11, Japan
(Received 25 October 1977; accepted for publication 20 December 1977)

This letter reports an optical waveguide TE-TM mode splitter using coupling between nonidentical waveguides. The experimental mode splitter was fabricated from glass compositions, and TE_0-TM_0 mode splitting has been demonstrated.

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An optical waveguide TE-TM mode splitter is a fundamental component in integrated optics. Two types of mode splitter have been reported so far: (1) the directional coupler type made of anisotropic material[1] and (2) the hybrid waveguide type using interference of a periodic multilayer.[2]

This letter describes a new type of mode splitter which makes use of coupling between nonidentical isotropic waveguides. A novel type of TE-TM mode filter based on the coupling between nonidentical waveguides has already been proposed by the present authors.[3] It is expected that the mode splitter can also be designed in essentially the same way as the mode filter. As in the case of the mode filter, we consider waveguide coupling in a thin-film multilayer structure, schematically shown in Fig. 1. Waveguide 1 is formed from substrate (refractive index n_s), guide layer 1 (refractive index n_1, thickness W_1), and an intermediate layer (refractive index n_I, thickness W_I). Waveguide 2 is formed from the intermediate layer, guide layer 2 (refractive index n_2, thickness W_2), and surrounding medium, i.e., air (refractive index n_a). Discussion is confined to the lowest-order modes, i.e., both TE_0 and TM_0 modes, for design simplicity.

According to the coupled-mode theory, normalized optical powers carried by waveguides 1 and 2 are given by[4]

$$P_{1m} = \frac{\Delta\beta_m^2}{\Delta\beta_m^2 + k_m^2} + \frac{k_m^2}{\Delta\beta_m^2 + k_m^2}\cos^2(\Delta\beta_m^2 + k_m^2)^{1/2}z,$$

$$P_{2m} = \frac{k_m^2}{\Delta\beta_m^2 + k_m^2}\sin^2(\Delta\beta_m^2 + k_m^2)^{1/2}z, \qquad (1)$$

$$\Delta\beta_m = \tfrac{1}{2}(\beta_{1m} - \beta_{2m}) + (c_{1m} - c_{2m}),$$

where m is either E or M for the TE or TM mode, respectively, $\Delta\beta_m$ is a propagation constant mismatch, β_{1m} and β_{2m} are the propagation constants of waveguides 1 and 2, respectively, in the absence of coupling (i.e., intermediate layer thickness $W_I \rightarrow \infty$), c_{1m} and c_{2m} are small corrections to β_{1m} and β_{2m}, respectively, caused by coupling between the waveguides, and k_m is coupling coefficient. Terms c_{1m} and c_{2m} are neglected in the following discussion for simplicity.

Mode characteristics can be calculated from the

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